LEIC-T 2025/2026
Aprendizagem - Machine Learning
Homework I
Deadline 1/10/2025 23:00
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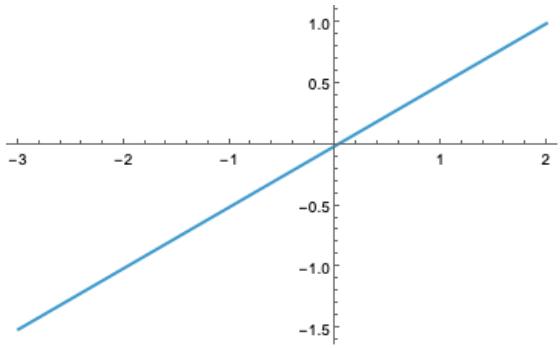
I) Correlation (3 pts)

Compute the correlation (Pearson correlation) and **Spearman's rank** for two variables $\mathbf{x}_1$ and $\mathbf{x}_2$=f($\mathbf{x}_1$). Are the values the same or different, please justify. (Indicate all computational steps!)

(a) (1 pts)

$\mathbf{x}_1$ =(−3, −2, −1,0,1,2)

for linear function



f($\mathbf{x}_1$)=0.5* $\mathbf{x}_1$
$\mathbf{x}_2$=f($\mathbf{x}_1$).

Solution

(if you assume it is a population you will get the same results for correlation):
Mean(x1)= -0.5, Mean(x2)= -0.25
Covariance for Sample cov(x1,x2)= 1.75
Standard Deviation for sample: s1= $\sqrt{\frac{7}{2}}$, *s2=* 0.9354
*corr(x1,x2)=cov(x1,x2)/(s1*s2)=**1***

R1=(1,2,3,4,5,6)
R2=(1,2,3,4,5,6)

Mean(R1)=3.5, Mean(R2)=3.5
Covariance for Sample *cov(R1,R2)=3.5*
Standard Deviation for sample: *s1=Sqrt(3.5), s2=Sqrt(3.5)*
*corr(R1,R2)=cov(R1,R2)/(s1*s2)=**1***

They are the same since the points are on the line.

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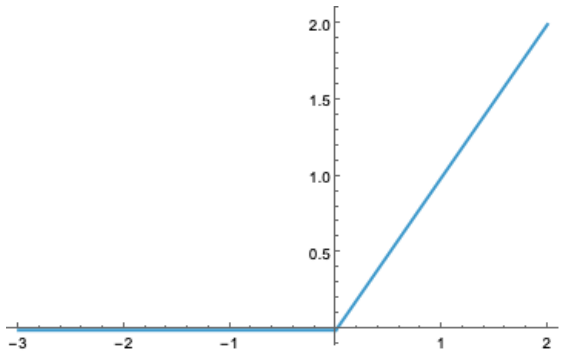
(b) (1pts)
$\mathbf{x}_1 = (-3, -2, -1, 0, 1, 2)$

for ReLU function



$f(\mathbf{x}_1)=\max(0, \mathbf{x}_1)$
$\mathbf{x}_2=f(\mathbf{x}_1)$.

Solution

(if you assume it is a population you will get the same results for correlation):
Mean(x1)=-0.5, Mean(x2)=0.5
Covariance for Sample cov(x1,x2)=1.3
Standard Deviation for sample: s1= 1.8708, *s2=* 0.8367
*corr(x1,x2)=cov(x1,x2)/(s1*s2)=* 0.83054

R1=(1,2,3,4,5,6)
R2=(2.5,2.5,2.5,2.5,5,6)

Mean(R1)=3.5, Mean(R2)=3.5
Covariance for Sample *cov(R1,R2)=* 2.5
Standard Deviation for sample: *s1=* 1.8708, *s2=* 1.5811
*corr(R1,R2)=cov(R1,R2)/(s1*s2)=* **0.8452**

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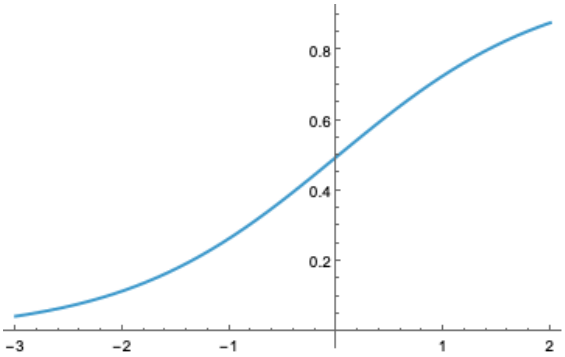
(c) (1pts)
$\mathbf{x}_1 = (-3, -2, -1, 0, 1, 2)$
for sigmoid also called logistic function



$f(\mathbf{x}_1)=\frac{1}{1+e^{-x1}}$ =1/(1+Exp(-$\mathbf{x}_1$)
$\mathbf{x}_2$=f($\mathbf{x}_1$).

Solution

(if you assume it is a population you will get the same results for correlation):
Mean(x1)=-0.5, Mean(x2)= 0.4245701
Covariance for Sample cov(x1,x2)= 0.623348
Standard Deviation for sample: s1=Sqrt(3.5), s2= 0.33689
*corr(x1,x2)=cov(x1,x2)/(s1*s2)=* 0.9891

R1=(1,2,3,4,5,6)
R2=(1,2,3,4,5,6)

Mean(R1)=3.5, Mean(R2)=3.5
Covariance for Sample *cov(R1,R2)=3.5*
Standard Deviation for sample: *s1=Sqrt(3.5), s2=Sqrt(3.5)*
*corr(R1,R2)=cov(R1,R2)/(s1*s2)=**1***

They are not the same since the ordered function is a line

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II) Decision Trees (4 pts)

Training Set:

Exam	*Weather*	*Summer*	*What to Do?*
No	*Sunny*	*Yes*	*Go to the beach*
Yes	*Sunny*	*Yes*	*Learn*
No	*Rain*	*No*	*Reading*
No	*Claudy*	*No*	*Reading*
Yes	*Claudy*	*Yes*	*Learn*
No	*Rain*	*Yes*	*Reading*
No	*Sunny*	*No*	*Go to the beach*
Yes	*Claudy*	*Yes*	*Learn*

(a) (1.5 pts)
Determine the root of decision tree using the ID3 algorithm with the target "*What to Do?*". Indicate the calculation. (Indicate all computational steps!)
Solution
p(B)=2/8=1/4, p(L)=3/8, p(R)=3/8
Log2[x]= Log[x]/Log[2]

I(table)= $-1/4 * \text{Log2}[1/4] - 3/8 * \text{Log2}[3/8] - 3/8 * \text{Log2}[3/8] = 1.561278$ ***bit***

$$E(P) = \sum_{i=1}^{n} \frac{|C_i|}{|C|} I(C_i) \qquad gain(P) = I(C) - E(P)$$

Exam: *Cy=(L,L,L) Cn=(B,B,R,R,R)*
*I(Cy)= -3/3*Log2[3/3]=-Log2[1] = Log2[1] =0 bit*
I(Cn)= $-2/5 * \text{Log2}[2/5] - 3/5 * \text{Log2}[3/5] = 0.97095$ *bit*

*E(Exam)= 3/8*0+5/8**$0.97095 = 0.606844$ *bit*
Gain(Exam)$= \mathbf{1.561278 - 0.606844 = 0.954434}$ ***bits***

Weather: *Cs=(B,L,B) Cr=(R,R) Cc=(R,L,L)*
I(Cs=$-1/3 * \text{Log2}[1/3] - 2/3 * \text{Log2}[2/3] = 0.918296$ *bit*
I(Cr)=-1 Log2[1]=0 bit*
I(Cc)= $-1/3 * \text{Log2}[1/3] - 2/3 * \text{Log2}[2/3] = 0.918296$ *bit*

E(Weather)=3/8*0.918296 + 2/8*0+3/8*0.918296=0.68872 bit
Gain(Weather)= $1.561278 - 0.68872 = 0.87256$ *bit*

Summer: *Cy=(B,R,L,L,L) Cn=(R,R,B)*
*I(Cy)= -1/5*Log2[1/5] -1/5*Log2[1/5] – 3/5*Log2[3/5]=* 1.37095 *bit*
*I(Cn)= -2/3*Log2[2/3] – 1/3*Log2[1/3]=* 0.918296 *bit*

E(Summer) $= 5/8 * 1.37095 + 3/8 * 0.918296$=1.20120475 bit
Gain(Summer)= $1.561278 - 1.20120475 = 0.36007$ *bit*

We chose ***Exam*** *as the root*

(b) (1.5 pts)
Determine the decision tree using the ID3 algorithm with the target "*What to Do?*"
Draw your decision tree.

~~*Exam*~~	*Weather*	*Summer*	*What to Do?*
~~*Yes*~~	*Sunny*	*Yes*	*Learn*
~~*Yes*~~	*Claudy*	*Yes*	*Learn*
~~*Yes*~~	*Claudy*	*Yes*	*Learn*

Ok, here we do not need to do anything.

In the second table we do the recursion

~~*Exam*~~	*Weather*	*Summer*	*What to Do?*
~~*No*~~	*Sunny*	*Yes*	*Go to the beach*
~~*No*~~	*Rain*	*No*	*Reading*
~~*No*~~	*Claudy*	*No*	*Reading*
~~*No*~~	*Rain*	*Yes*	*Reading*
~~*No*~~	*Sunny*	*No*	*Go to the beach*

p(B)=2/5, p(R)=3/5
I(table)= $-2/5 * \mathrm{Log2}[2/5] - 3/5 * \mathrm{Log2}[3/5] = 0.97095$ ***bit***

Weather: *Cs=(B,B) Cr=(R,R) Cc=(R)*
I(Cs)=0 bit
I(Cr)=0 bit
I(Cc)=0 bit

E(Weather)=0 bit
Gain(Weather)= $\mathbf{0.97095}$ ***bit***

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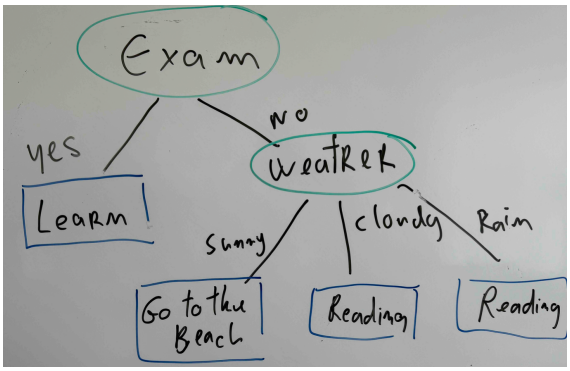
Summer: *Cy=(B,R) Cn=(R,R,B)*
I(Cy)= 1 bit
*I(Cn)= -2/3*Log2[2/3] – 1/3*Log2[1/3]=* 0.918296 *bit*
E(Summer) $= 2/5 * 1 + 3/5 * 0.918296$=0.9509776 bit
Gain(Summer)= 0.97095-0.9509776 = 0.01997 *bit*

*We chose **Weather** as the root of the sub-tree*



Draw confusion matrix for the test set

Test Set:

Exam	*Weather*	*Summer*	*What to Do? (True according to the test set)*
No	*Sunny*	*Yes*	*Go to the beach*
Yes	*Sunny*	*Yes*	*Go to the beach*
No	*Rain*	*No*	*Reading*
No	*Claudy*	*No*	*Go to the beach*
Yes	*Claudy*	*Yes*	*Go to the beach*
No	*Rain*	*Yes*	*Learn*
Yes	*Sunny*	*No*	*Go to the beach*
Yes	*Claudy*	*Yes*	*Go to the beach*

Exam	*Weather*	*Summer*	*True*	*Predicted (as trained)*
No	*Sunny*	*Yes*	*Go to the beach*	*Go to the beach*
Yes	*Sunny*	*Yes*	*Go to the beach*	*Learn*
No	*Rain*	*No*	*Reading*	*Reading*
No	*Claudy*	*No*	*Go to the beach*	*Reading*
Yes	*Claudy*	*Yes*	*Go to the beach*	*Learnig*
No	*Rain*	*Yes*	*Learn*	*Reading*
Yes	*Sunny*	*No*	*Go to the beach*	*Learn*
Yes	*Claudy*	*Yes*	*Go to the beach*	*Learn*

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			TRUE	
		Learn	Got to the beach	Reading
	Learn	0	4	0
PREDICTED	Got to the beach	0	1	0
	Reading	1	1	1

or transpse

			PREDICTED	
		Learn	Got to the beach	Reading
	Learn	0	0	1
TRUE	Got to the beach	4	1	1
	Reading	0	0	1

III Software Experiments (2pts)

Imagine you are asking ChatGPT a question to generate a python program of the ID3 example of the theoretical lectures. What would be the question?

"Can you indicate me ID3 algorithm in python using Shannon's entropy on a business application?"

Solution

This is cool, but it helps **a lot to know the theory behind the algorithm** to understand the Python code. As you see it uses with *high probabil*ity the same example from the book

"George F Luger, Artificial Intelligence: Structures and Strategies for Complex Problem Solving 6th, Edition, Pearson, 2008"

On which it was trained. By interaction with ChatGPT you can adapt your algorithm to your required application. Like for example reading a special data format ☺

Download the jupyter notbook HM1_DT_25.ipynb.

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(a) (1pts)
Split the data using the command (in the notebook)

X_train, X_test, y_train, y_test = train_test_split(X, y, train_size=**value**,stratify=y,random_state=**your_group number**)

Partition data with the *train_test*_split values *0.1, 0.9* and indicate the depth and the accuracy of each the decision tree (if you have no group yet, put the last three digits of student nr).

Why is the depth of the decision tree bigger and the accuracy higher for the *train_test_split* values *0.9* than for the value *0.1*?

(b) (1pts)
Now perform the same experiment with the *train_test_split* value 0.9 without the command *stratify=y*, see

X_train, X_test, y_train, y_test = train_test_split(X, y, train_size=**0.9**, ,random_state=**your_group number**)

Why is now the accuracy lower? Please write only one/two sentence/s… Hint, look in the scikit-learn documentation for meaning of the command *stratify=y.*

Solution

See the file HS1_DecsionTrees_25.ipynb